

ENGINEERING CHEMISTRY
SUBJECT CODE- NBS-102/NBS-202

COURSE OBJECTIVES:

1. To enable the students to understand about the Chemistry of Atomic and Molecular structure, Chemistry of advanced Materials like Liquid crystals, Graphite & fullerenes.
2. To enable the students to understand and apply the concepts related to Electrochemistry, Batteries, Corrosion and Chemistry of Engineering Materials like cement and POP.
3. To enable the students to understand and apply spectroscopic techniques, principles of green chemistry and its environmental impacts, Analysis of coal & determination of calorific values.
4. To enable the students to understand and apply detailed concepts of water source, water impurities, hardness of water, Phase rule and Lubricants
5. To enable the students to understand detailed concepts related to polymers, Polymerization, and some well-known name reaction with mechanism.

SYLLABUS:

Content	Contact Hours
Unit-1: Chemical Bonding	8
Molecular orbital theory and its application to homo and hetero nuclear diatomic molecules and ions B ₂ , N ₂ , O ₂ , F ₂ , CO, NO, HF, O ₂ ⁺ , O ₂ ⁻ , O ₂ ²⁺ , Band theory -conductor, insulator and Semiconductor. Liquid Crystals its types and application Imperfection in ionic crystalline solids Conductor, Semiconductor, insulators	
Unit-2: Electrochemistry and Engineering Materials:	8
Basic concepts of electrochemistry, Galvanic cell, Nernst equation, Faraday's law and Numerical. Corrosion: Introduction to corrosion, Types of corrosion, Cause of corrosion, Corrosion prevention and control. Cement: Constituents, manufacturing, hardening and setting, hydraulic and Non hydraulic cement, Portland cement.	
Unit-3: Green Chemistry and Characterization	8
Introduction, 12 principles and importance of green Synthesis, Green Chemicals and Solvents, Synthesis of typical organic compounds by conventional and Green route (Adipic acid and Paracetamol), Environmental impact of Green chemistry on society. Spectroscopic Techniques and Applications: Elementary idea and simple applications of UV, IR, Mass and NMR, Numerical problems. Fuels Classification of Fuels, Analysis of Coal and Calorific value (Bomb calorimeter).	
Unit-4: Water and Phase Rule	8
Water Characteristics of water, Soft and Hard of water (with types), Techniques for water softening (Lime-Soda, Zeolite, Ion Exchange and Reverse Osmosis process), Degree of Hardness, Numerical problems. Phase Rule Derivation of Phase rule and its application to one component water and sulphur system. Lubricants Definition, Functions, Mechanism and Classification of Lubricants	

Unit-5: Chemistry of Materials	8
Polymers: Classification, types of Polymerization and its mechanism, Thermosetting, Thermoplastic, HDP, LDP, Conducting and Biodegradable polymers, Ziegler-Natta Polymerisation, Preparation, Properties, Industrial applications of Bakelite, Natural and synthetic rubber, Nylon66, Buna-N and Buna-S. Organometallic Compounds: Preparation and applications of RMgX and LiAlH ₄ . Name Reactions SN1 and SN2, Aldol and cross Aldol reaction, Reimer-Tiemann reaction, Friedel-Crafts Alkylation and Acylation	

Course Outcome (CO):

On completion of course the students are able:

CO	CO Statement	Bloom's Level
CO1	Get an understanding of the theoretical principles of chemistry of molecular structure, bonding and properties, Chemistry of liquid crystals, graphite and fullerenes, and defects occurring in ionic solids	Understanding, Apply
CO2	To understand the concept of current in a cell/ battery and use of electrodes, causes and prevention of corrosion, manufacturing of cement and POP.	Understanding, Analyze
CO3	To understand the green approaches of eco sustainable chemistry and modern characterizations of synthesized compounds	Understanding, Apply
CO4	Develop understanding of the sources, impurities and hardness of water and removal techniques.	Understanding, Analyze
CO5	To know the synthesis, applications and mechanism of industrially important polymers	Understanding

Reference Books:

1. Engineering Chemistry by Sunita Ratan, 1st Edition, S. K. Kataria and Sons New Delhi.
2. Engineering Chemistry by SS Dara, S Chand & Co Ltd
3. Engineering Chemistry by Jain & Jain, S.Chand & Comp, New Delhi
4. Engineering Chemistry by Gurdeep Raj, S. K. Kataria and Sons New Delhi.
5. Engineering Chemistry by OG Palanna, Mc Graw Hill Education, New Delhi
6. Engineering Chemistry by Shashi Chawala, Dhanpat Rai Publishing Comp, New Delhi
7. University Chemistry by BH Mahan
8. University Chemistry by CNR Rao